MIL-STD-1328 B

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MILITARY STANDARD

COUPLERS, DIRECTIONAL (COAXIAL LINE, WAVEGUIDE, AND PRINTED CIRCUIT), SELECTION OF

FSC 5985

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DEPARTMENT OF DEFENSE
Washington, D.C. 20301

Couplers, Directional (Coaxial Line, Waveguide, and Printed Circuit), Selection of

MIL-STD-1328B

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard establishes requirements for the selection of directional couplers for use in military equipments.

1.2 Purpose. The purpose of this standard is to:

a. Provide the equipment designer with a list of directional couplers considered standard for use in military applications.
b. Restrict the number of directional couplers for use in military applications in order to provide effective logistic support of equipment.
c. Establish criteria pertinent to choice and application of directional couplers in military equipment.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of this standard to the extent specified herein.

SPECIFICATION

MILITARY

MIL-C-15370 - Couplers, Directional (Coaxial Line, Waveguide, or Printed Circuit), General Specification For.

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 The terms used in this standard are those commonly encountered in directional coupler engineering practice.

4. GENERAL REQUIREMENTS

4.1 Selection of directional couplers. Directional couplers to be used in military equipment and systems shall be selected from those listed in table I.

4.2 Criteria for inclusion. The criteria for the selection of directional couplers for this list are:

a. The coupler shall be considered by representatives of the military departments the best available type for current application.
b. Availability of the coupler shall be reasonably certain.
c. The coupler shall have an approved military specification.

4.3 Application and use. Directional couplers used in military equipment shall be from lots possessing acceptable material and physical and electrical characteristics and shall in no manner degrade the operational characteristics of the equipments in which used.

4.4 Detailed requirements. The detailed requirements for directional couplers listed in this standard are covered by the applicable MIL-C-15370 specification sheet.

5. DETAILED REQUIREMENTS (Not applicable)

6. NOTES (Not applicable)

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TABLE I. Listing of directional couplers.

1/ Part number	Frequency range (GHz)		Equivalent transmission line	Coupling factor (nominal) (dB)
M15370/1-001	.120 thru	.240	RG-92/U	20
1-002	.240	.480	92	20
1-003	.480	.960	92	20
1-004	.960	1.990	92	20
1-005	1.990	4.000	92	20
2-001	2.650	5.350	48	20
2-002	5.350	4.000	48	20
2-003	3.950	5.000	49	20
2-004	5.000	6.000	49	20
2-005	5.300	7.000	50	20
2-006	7.000	8.200	50	20
2-007	7.000	8.500	51	20
2-008	8.500	10.000	51	20
2-009	12.400	15.200	91	20
2-010	18.000	22.200	53	20
2-011	22.000	26.500	53	20
2-012	26.500	31.500	96	20
2-013	31.500	36.000	96	20
3-001	.225	.460	Series N coaxial	10
3-002	.950	2.000	Series N coaxial	10
3-003	2	4	Series N coaxial	10
3-004	4	10	Series N coaxial	10
3-005	.216	.450	Series N coaxial	20
3-006	.225	.460	Series N coaxial	20
3-007	.940	1.975	Series N coaxial	20
3-008	.950	2.000	Series N coaxial	20
3-009	1.9	4.0	Series N coaxial	20
3-010	4	10	Series N coaxial	20
3-011	.225	.460	Series N coaxial	30
3-012	.950	2.000	Series N coaxial	30
3-013	2	4	Series N coaxial	30
3-014	4	10	Series N coaxial	30
3-015	.030	.960	Series N coaxial	6 dB/octave
3-016	.1	1.2	Series N coaxial	6 dB/octave
4-001	5.85	8.20	RG-50/U	3
4-002	5.85	8.20	50	10
4-003	5.85	8.20	50	20
4-004	7.05	10.00	51	3
4-005	7.05	10.00	51	10
4-006	7.05	10.00	51	20
4-007	8.20	12.40	52	3
4-008	8.20	12.40	52	10
4-009	8.20	12.40	52	20
4-010	12.40	18.00	91	20
4-011	18.00	26.50	66	3
4-012	18.00	26.50	66	10
4-013	18.00	26.50	66	20
5-001	5.95	5.85	49	20
5-002	8.20	12.40	52	20
6-001	5.50	9.60	67	20
6-002	8.20	12.40	67	40
6-003	9.00	9.20	67	40

See footnote at end of table.

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TABLE I. Listing of directional couplers - Continued

1/ Part number	Frequency range		Equivalent transmission line	Coupling factor (nominal)
				(dB)
M15370/9-004	1.0 thru	2.0 GHz	SMA coaxial	30
9-006	2.0	4.0		10
9-007	2.0	4.0		20
9-008	2.0	4.0		30
9-010	4.0	8.0		10
9-011	4.0	8.0		20
9-012	7.0	12.4		6
9-013	7.0	12.4		10
9-014	7.0	12.4		20
9-016	12.4	18.0		10
9-017	12.4	18.0		20
9-018	7.0	12.4		30
9-019	11.0	17.0		10
9-020	11.0	17.0		20
9-021	11.0	17.0		30
9-022	7.5	16.0		6
9-023	7.5	16.0		10
9-024	7.5	16.0		20
9-025	7.5	16.0		30
9-026	1.0	18.0		16
10-001	.50	1.0		6
10-002	.50	1.0		10
10-003	.50	1.0		20
10-004	1.0	2.0		6
10-005	1.0	2.0		10
10-006	1.0	2.0		20
10-007	2.0	4.0		6
10-008	4.0	8.0		6
10-009	8.0	12.4		10
10-010	8.0	12.4		20
10-011	12.4	18.0		6
11-001	0.1	2.0	N coaxial	20
11-002	4.0	8.0	N coaxial	20
14-001	.025	8.0	N and C	40
15-001	7.0	11.0	RG-320/U	50 and 30
15-002	10.8	18.0	RG-349/U	50 and 30
16-001	25	35 MHz	SMA coaxial	25
16-002	125	250		30
16-003	190	210		10
16-004	250	500		6
16-005	250	500		10
16-006	250	500		20
16-007	250	500		30
16-008	285	315		25
17-001	60	80		10

See footnote at end of table.

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TABLE I. Listing of directional couplers - Continued.

1/ Part number	Frequency range	Equivalent transmission line	Coupling factor (nominal)
			(dB)
M15370/18-001	0.2 thru 250 MHz	Printed circuit	19.5
18-002	0.5 500		11.5
18-003	1.0 1000		11.0
18-004	10.0 400		10.0
18-005	50.0 400		10.0
18-006	50.0 400		15.0
18-007	50.0 400		20.0
18-008	250 1000		10.0

1/ Applicable AN nomenclature is listed in table II.

TABLE II. Cross-reference of AN nomenclature to part number.

AN nomenclature	Part number	AN nomenclature	Part number
CG-176/AP	M15370/6-001	CU-1511/U	M15370/4-002
CU-187/U	1-001	CU-1512/U	4-003
CU-188/U	1-002	CU-1513/U	4-007
CU-189/U	1-003	CU-1514/U	4-008
CU-190/U	1-004	CU-1515/U	4-009
CU-191/U	1-005	CU-1516/U	3-005
CU-198/U	2-001	CU-1517/U	3-007
CU-199/U	2-002	CU-1518/U	3-009
CU-200/U	2-003	CU-1519/U	4-010
CU-201/U	2-004	CU-1520/U	3-016
CU-202/U	2-005	CU-1521/U	3-015
CU-203/U	2-006	CU-1522/U	4-011
CU-204/U	2-007	CU-1523/U	4-012
CU-205/U	2-008	CU-1524/U	4-013
CU-208/U	2-009	CU-1525/U	3-001
CU-210/U	2-010	CU-1526/U	3-006
CU-211/U	2-011	CU-1527/U	3-011
CU-212/U	2-012	CU-1528/U	3-002
CU-213/U	2-013	CU-1529/U	3-008
CU-988/U	5-002	CU-1530/U	3-012
CU-1506/U	5-001	CU-1531/U	3-003
CU-1507/U	4-004	CU-1532/U	3-013
CU-1508/U	4-005	CU-1533/U	3-004
CU-1509/U	4-006	CU-1534/U	3-010
CU-1510/U	4-001	CU-1535/U	3-014

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